11.1.2 Sum of digits using recursion

ALGORITHM

Step 1: Start

Step 2: Input an integer n

Step 3: Call function sum(n)

Step 4: If n == 0, return 0

Step 5: Else compute:

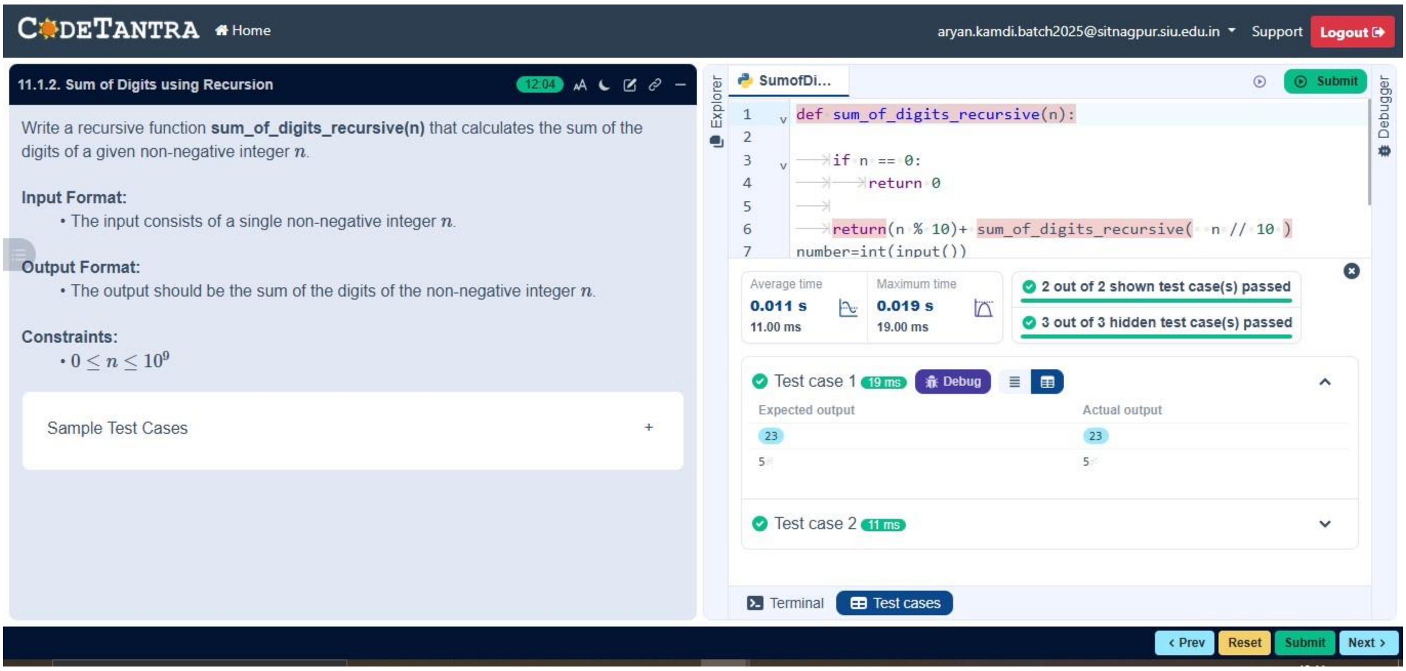
  a) r = n % 10 (last digit)

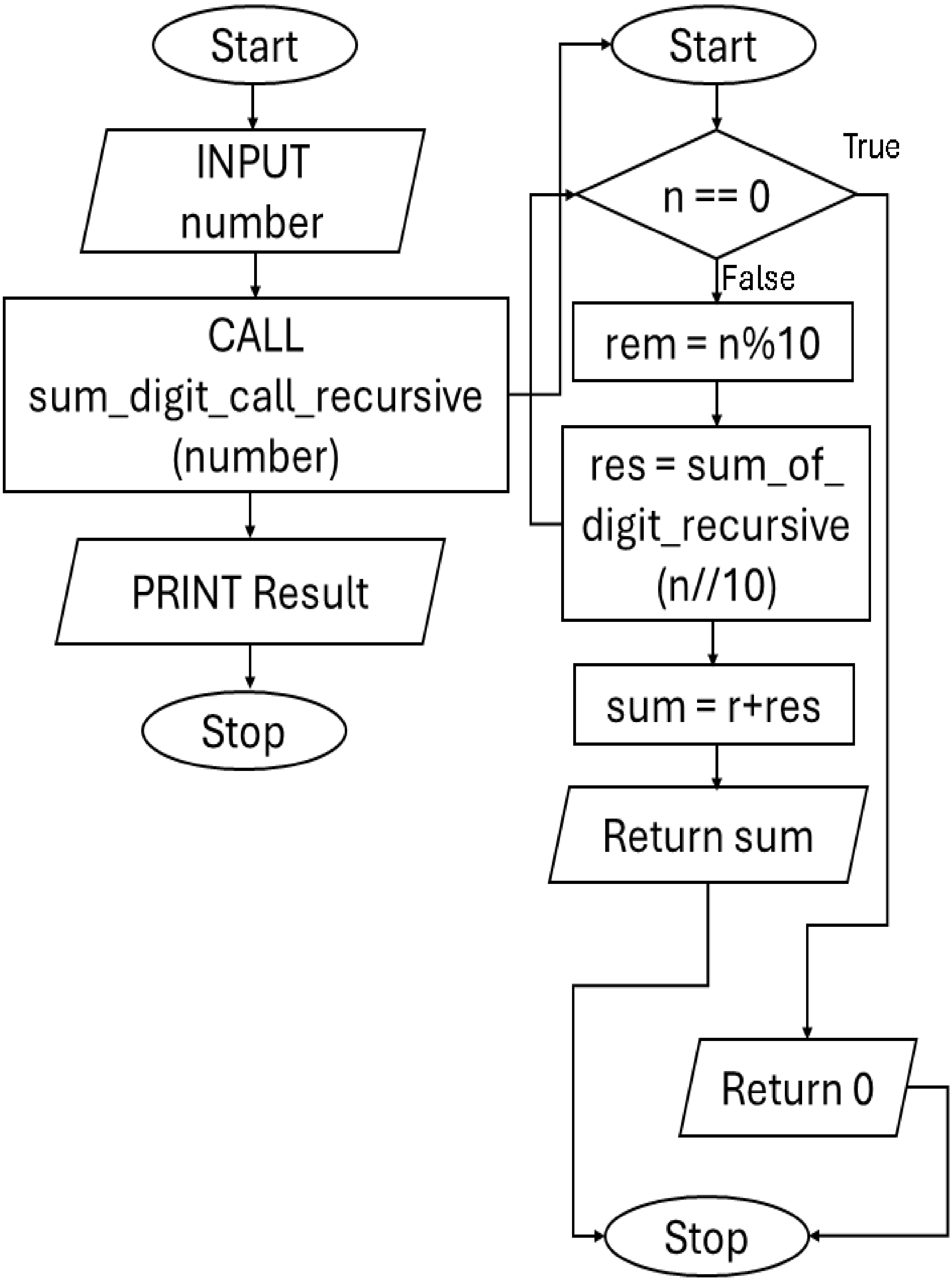
  b) q = n // 10 (remaining number)

Step 6: Return r + sum(q)

Step 7: Print the result

Step 8: Stop





Start
INPUT number
CALL sum_digit_call_recursive (number)
PRINT Result
Stop
Start
n == 0
True
False
rem = n%10
res = sum_of_ digit_recursive (n//10)
sum = r+res
Return sum
Return 0
Stop